INSTALLATION INSTRUCTIONS

Power Glove Vision Installation Guide

Install, pair, and play with the PowerGlove Vision Controller and RetroPie.

2026 EDITION / IAIN BENNETT / MIT LICENSE

Install PowerGlove Vision with one script on the **PowerGlove Vision Controller (Arduino UNO Q)** and one on RetroPie. The scripts prepare the software and startup helpers; you finish by pairing the devices, positioning the camera, and testing a game.

Choose **Setup -> Matrix attract mode** to keep the idle animation On, Dim it, or turn it Off except for faint connection pixels. This does not change game displays, T, L, or gesture recognition. The setting saves without a tracker restart.

For an existing installation, this update changes controller transport on both computers. Stop controller output, update both to matching software, then start and test input. Mixed old/new versions do not deliver input with the default settings. See signed controller transport and upgrades for staged upgrades and rollback.

Install this release candidate

To test **v0.3.2-rc.6**, close games and stop controller output, then run the matching command on each device. These explicit commands select the candidate; the normal commands later in this guide select the latest stable release.

On the PowerGlove Vision Controller:

```sh
curl -fLO \
  https://github.com/mathan416/PowerGlove-Vision/releases/download/v0.3.2-rc.6/install-uno-q.sh \
  && bash install-uno-q.sh --version v0.3.2-rc.6
```

On RetroPie:

```sh
curl -fLO \
  https://github.com/mathan416/PowerGlove-Vision/releases/download/v0.3.2-rc.6/install-retropie.sh \
  && bash install-retropie.sh --version v0.3.2-rc.6
```

Verify both report the same candidate, then follow the pairing/first-game checks below. Existing hand settings and pairing files are preserved. The Controller installer also updates the matrix firmware. Review coordinated transport upgrades and rollback before replacing an older installation. Candidate testing still needs live gameplay, stationary-jitter, and camera-to-display latency checks; see the changelog.

1. Prepare your devices

You need a provisioned PowerGlove Vision Controller, a working RetroPie system, a UVC USB camera, a powered USB hub, and a physical controller for RetroArch setup. Put both devices on the same trusted local network with internet access. Supply your own games; no ROMs or BIOS files are included.

For a new Controller, use Arduino App Lab to complete board setup and networking. Record both devices' hostnames. Connect the camera through the powered hub. You do not need to import PowerGlove Vision through App Lab or build a ZIP on your computer. The installer builds and uploads the Arduino sketch for you.

Open a terminal on each device, either locally or over SSH. For the Controller:

```sh
ssh arduino@UNO-Q-NAME.local
```

Replace the example hostnames with your actual names. Use your normal account; the scripts request your sudo password when administrator access is needed. They do not store it. Close games before installing. Leave the devices powered and connected while installation runs; first setup can take several minutes.

The commands below automatically select the latest published stable release from GitHub. You do not need to enter a release tag or set shell variables. Run both installers during the same setup session and compare the reported release names; if a new release appeared between runs, rerun the older installation. Development prereleases are available separately in the technical reference. The latest release must include the installer assets before these commands work.

2. Run the PowerGlove Vision Controller installer

Run this single line in the Controller terminal:

```sh
SH
curl -fLO \
  https://github.com/mathan416/PowerGlove-Vision/releases/latest/download/install-uno-q.sh \
  && bash install-uno-q.sh
```

The script verifies its download, installs the app and sketch, and configures automatic startup. It includes the early-start hourglass helper and the Shutdown button's system helper. No separate helper commands are needed. Existing pairing, calibration, and personal tuning are preserved when updating. The supplied `config/profiles.json` baseline is backed up and replaced so current shared recognition defaults take effect. That baseline includes the tested movement hysteresis, native full-field mapping, and low-lag stabilization. The installer does not copy a maintainer's neutral-hand coordinates: those measurements depend on each camera, distance, and playing position.

If the script reports a failure, stop and follow its message. If `curl` is missing, install it with `sudo apt-get install curl ca-certificates`, then retry. The installer checks compatibility before changing the application.

Checkpoint: Open `http://UNO-Q-NAME.local:8088/dashboard` in your browser. Dashboard should load. With gestures off, a closed camera is normal. Open **Play** or **Glove Academy** to check that your camera view and whole hand appear, then return to Dashboard with controller transmission stopped.

3. Run the RetroPie installer

Run this single line in the RetroPie terminal:

```sh
SH
curl -fLO \
  https://github.com/mathan416/PowerGlove-Vision/releases/latest/download/install-retropie.sh \
  && bash install-retropie.sh
```

For a new installation, the script asks for your Controller hostname or IP address. There are no placeholders to replace in the command.

The script installs the receiver, controller mapping, game-launch integration, and automatic startup. Existing cabinet hooks and controller assignments remain. It offers missing emulator installation through RetroPie Setup and checks registered games, including Bad Street Brawler's Glove Zap configuration.

Follow any emulator or game ACTION messages. Missing games do not prevent the base installation. If asked to launch and exit FCEUmm once, do that and rerun the installer. If you use another emulator, the installer asks before selecting FCEUmm for Bad Street Brawler.

If a registered Super Glove Ball ROM is present, the installer also offers the optional `lr-nestopia-powerglove` core. Accepting installs Git and the standard build tools, downloads the pinned GPLv2 Nestopia source, applies the included patch, builds on that RetroPie, and registers a second emulator entry. It does not change the ROM's saved emulator:

FCEUmm remains selected until you choose `lr-nestopia-powerglove` from RetroPie's per-ROM launch menu. Declining is safe and leaves the tested joystick fallback unchanged. The build needs internet access and may take several minutes; no ROM is read or copied by the build.

Checkpoint: The report confirms that receiver startup is configured. Pairing and live gameplay checks will still be listed as actions.

4. Pair the devices

Pairing gives both devices the same private token. Use the recommended one-time-code method after both installers finish.

1. On RetroPie, run `sudo /opt/powerglove/bin/powerglove-pair`. Leave the command running; it prints a 20-character code that expires after two minutes.
2. In your computer's browser, open `https://UNO-Q-NAME.local:8443/setup`. This is the secure Setup page; ordinary HTTP Setup cannot accept pairing credentials.
3. Enter your RetroPie hostname and its one-time code, then select **Prepare code pairing**.
4. Read the identifier after `ID` on the Controller's matrix. Compare it with the beginning of the browser certificate's SHA-256 fingerprint. The locally generated certificate may cause a browser warning; verify the fingerprint before continuing.
5. If the identifiers match, select the confirmation checkbox, enter the six-digit PIN shown after `PN` on the matrix, and select **Complete pairing**.
6. If Setup asks you to save the console destination, select **Save connection settings** above. On RetroPie, confirm that the helper reports completion and exits. Run `sudo systemctl status powerglove-receiver.service`; the receiver should be active.

If the code expires, restart the RetroPie command and prepare a new attempt. Never paste the private token into a document, screenshot, or support request.

Alternative: pair with your RetroPie password

Use this route only if RetroPie accepts SSH password login and your account can run `sudo` with that password.

1. Open secure Setup, expand **Pair using an SSH password**, and enter the RetroPie hostname and username.
2. Select **Prepare password pairing** and compare the matrix `ID` with the browser certificate fingerprint.
3. If they match, select the confirmation checkbox, enter the matrix PIN and your RetroPie password, and complete pairing.
4. Confirm that the receiver service is active on RetroPie.

The password is used for one SSH operation and is not stored. If neither pairing route works, follow the token-management reference.

Connection settings and recovery

Connection and startup saves the console address and startup game profile. Port, camera, and pairing-key replacement are under **Advanced connection settings**. **Check console address** only checks name resolution. If loading fails, use **Reload saved settings**; if a save fails, correct or retry it without losing fields. Start/Stop may show a pending request while tracking reconnects.

Connection saves restart tracking. The separate **Save attract mode** action changes only the idle matrix display. Hand setup, players, and backups are in **Glove Academy**. Existing private settings and calibration remain preserved through the normal installation/upgrade process; the independent Wi-Fi pixel needs the updated matrix firmware. Normal installation and Wi-Fi deployment also install its unprivileged host status sampler. The receiver timeout correction takes effect after updating RetroPie as well as the Controller application.

5. Calibrate and test a game

1. On Dashboard, select a profile, wait for the camera, and show your hand. On first use, the app collects a neutral reference automatically. Use **Calibrate** if your resting position produces unwanted movement or your camera/playing position changed. Hold a relaxed, open hand still at the intended center and distance until the button reports completion.
2. Select **Start controller**. This allows controller packets to reach RetroPie and creates the virtual input device.
3. On RetroPie, run `grep -A8 -B2 'PowerGlove Vision' /proc/bus/input/devices`. Look for the device name **PowerGlove Vision**. If it is missing, check pairing and the receiver service before changing emulator settings.
4. Use your physical controller to open RetroArch. Go to **Settings > Input > RetroPad Binds > Port 1 Controls** and select **PowerGlove Vision**. Menu labels can vary with the RetroArch version.
5. Check the D-pad, A, B, Start, and Select assignments. The installer provides an automatic mapping; adjust bindings only if needed, then save the controller profile or RetroArch configuration.
6. Test movement and buttons in a game. If your cabinet merges multiple controllers, also configure that merger to accept the virtual device.

Checkpoint: A gesture changes the intended control in the running game. Seeing the device name or a running service alone is not an end-to-end test. Bad Street Brawler's Glove Zap uses simultaneous Left + Right through the standard gamepad path. The RetroPie installer checks the game-specific FCEUmm option. Follow any remaining ACTION message and rerun the installer after resolving it. See Glove Zap setup. No extra-trigger assignment or receiver change is required.

A calibration uses 24 clear observations with at least 70% tracking confidence. Repeating it from the same position should give closely comparable center, scale, wrist, and jitter values, but natural landmark variation prevents an exact numeric match. Calibration is shared by every profile. Closing Dashboard after this check stops its 5 fps diagnostic preview work without stopping hand tracking or controller delivery.

For Super Glove Ball testing, enter RetroPie's launch menu while starting the ROM and choose either `lr-fceumm` or `lr-nestopia-powerglove`. FCEUmm uses the ordinary D-pad and buttons for the whole session. The native core uses absolute X/Y/Z plus open-hand, fist, and index-point packets. Full-game cabinet play has confirmed grab/throw, index fire, and fist-plus-forward Power Punch. Continuous movement is playable, with further latency refinement still planned. Wrist rotation and remaining unused native packet fields stay neutral. Shared recognition remains available to every FCEUmm profile. A per-ROM selection is remembered, so choose FCEUmm again whenever you want the complete fallback.

6. Confirm startup and finish

- Launch a registered game and check the expected profile on Dashboard and the matrix.
- Exit the game and confirm that gestures turn off.
- Reboot both devices normally. Confirm that Dashboard returns and the matrix progresses through startup to the selected mode. The early-start helper is enabled automatically for this boot.
- Check that your saved tuning and calibration remain available.
- Select **Start controller** when ready and verify movement and buttons in the game.

The installers never reboot or request a shutdown automatically. They check that the Controller shutdown helper is ready. The tested Arduino UNO Q hardware restarts after a halt; a disconnected website or blank matrix is not proof that power can be removed.

Updates and checks

Updates keep an inventory of installed application files. After the first inventory is created, unchanged obsolete files are backed up and removed; your own changes are kept and reported. If an update is interrupted, follow its recovery instructions before trying again.

To update, repeat the same single-line commands on both machines. Each selects the latest published stable release. Changed managed files are backed up, and the installer prints their location. It asks before interrupting an active Controller session. Close RetroArch before updating RetroPie. `config/profiles.json` is intentionally replaced; saved personal tuning remains in `data/gesture-tuning.json`.

For checks only, use the script you already downloaded:

```sh
SH
# On the PowerGlove Vision Controller:
bash install-uno-q.sh --check
# On RetroPie:
bash install-retropie.sh --check
```

Checks do not download, install, restart, or change anything. They may request sudo access to inspect protected settings.

Report	Meaning
PASS	The named check succeeded.
FAIL	Resolve the problem before proceeding, then rerun the installer or checks.
ACTION	Complete the named step, such as pairing, adding games, or checking gameplay.

A successful technical installation can still report ACTION for the physical checks. Those checks need you and your cabinet.

If movement feels delayed after installation, use the read-only latency baseline before changing sensitivity or video settings. It separates Controller timing from the receiver, emulator, and display checks and does not enable controls.

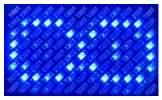
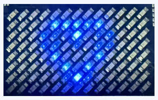
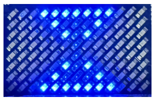
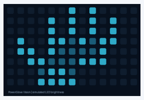
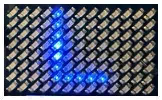
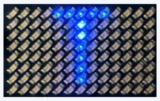
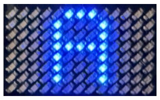
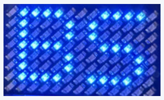
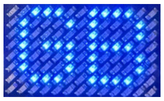
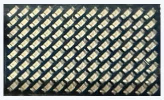
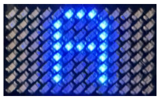
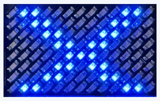
For a specific version or a development prerelease, use the technical installation reference. It also explains compatibility, package building, backups, and recovery.

If something does not work

- **Download fails:** confirm that the latest stable release includes installer assets, check internet access, and retry. A failed verification installs nothing.
- **Unsupported board software:** complete App Lab provisioning or use a compatible project release. Do not bypass the installer's compatibility check.
- **Website does not open:** try the Controller's current IP address instead of its hostname. Use HTTP on port 8088 and HTTPS on port 8443.
- **Camera missing:** open Glove Academy and wait for the camera view. The Controller host helper automatically enrolls the single UVC camera and its parent hub on first successful use, even if no camera was connected during installation. After enrollment it makes one guarded reset attempt during a sustained outage. If it remains missing, reconnect or power-cycle the camera and check the powered hub and cable; USB Ethernet may briefly disconnect during recovery.
- **No controller in RetroArch:** finish pairing, select Start controller, and select PowerGlove Vision for Port 1 using your physical controller.
- **Partial installation:** correct the reported problem and rerun the same release. Keep the printed backup location for recovery.

For diagnostic commands or manual repair, use the configuration reference.

Read the matrix and open the web pages

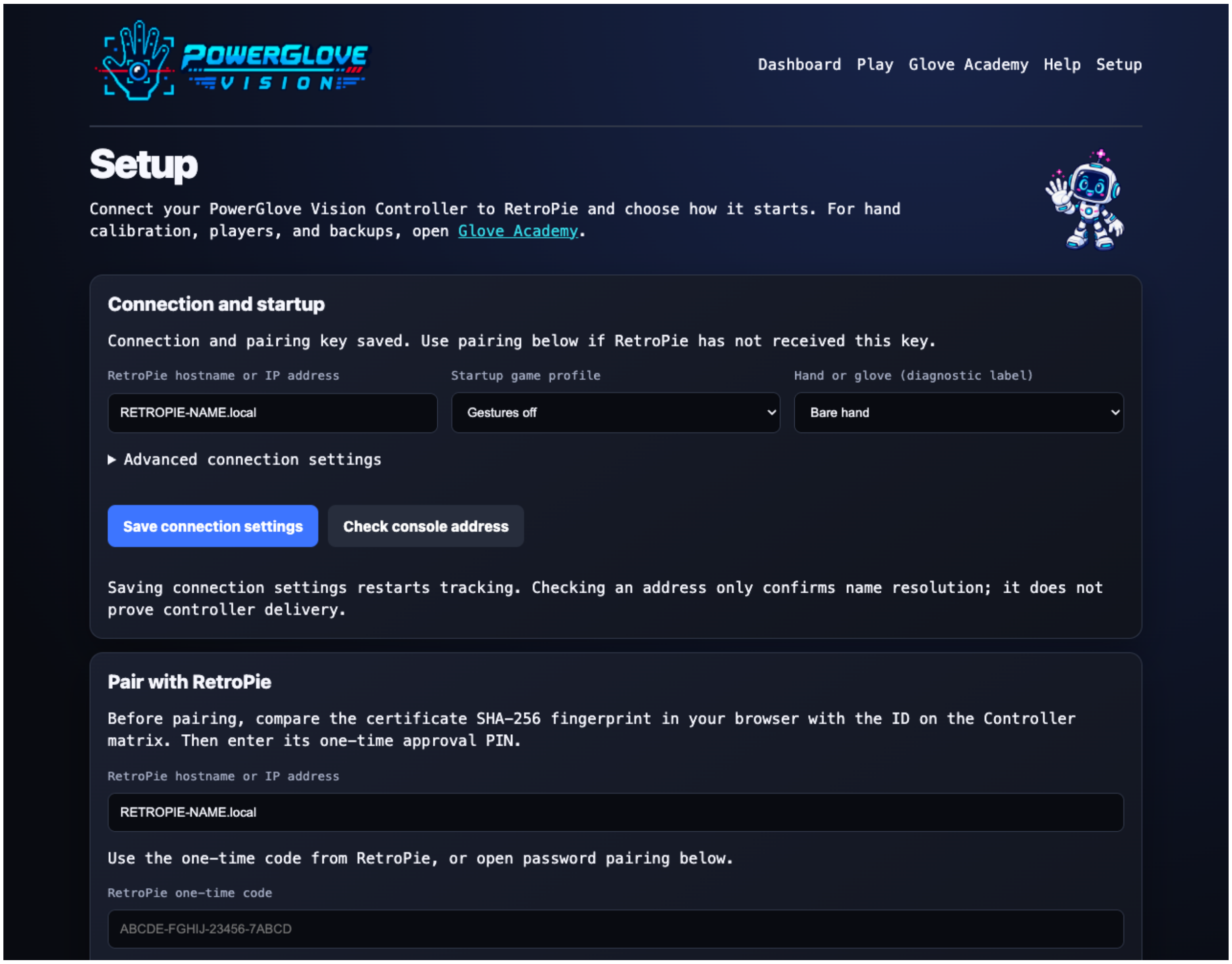
Display	See it	Meaning
Arduino boot logo		System startup, before the app display.
System heart		System startup is progressing.
Pulsing hourglass		PowerGlove Vision is starting.
Lightning and animated glove		Gestures are off. The revised animation requires updated matrix firmware.
Scanning `L`		Play or Glove Academy practice is active; controller output is paused.
Scanning `T`		Tune gestures is active; controller output is paused.
`A`-`I`		The corresponding profile is selected; Program A is shown.
`BS`		Bad Street Brawler is selected.
`GB`		Super Glove Ball is selected.
Blank matrix		No LEDs are illuminated. Check board power and Dashboard; blank does not confirm shutdown.
Pulsing profile code		A calibrated hand is being tracked. Confirm controller output separately.
Blinking X		The app has requested an error display. Check Dashboard for the cause.

See the Matrix display guide for the complete animations and startup sequence. An animation does not prove that shutdown has finished.

Page	Address
Dashboard	`http://UNO-Q-NAME.local:8088/dashboard`
Play	`http://UNO-Q-NAME.local:8088/play`
Glove Academy	`http://UNO-Q-NAME.local:8088/learn`
Games (lower Setup section)	`http://UNO-Q-NAME.local:8088/setup#games-section`
Help and printable manuals	`http://UNO-Q-NAME.local:8088/help`
Connection settings	`http://UNO-Q-NAME.local:8088/setup`
Secure pairing	`https://UNO-Q-NAME.local:8443/setup`

In Glove Academy, choose or add a player before practicing. Progress and hand sensitivity persist across restarts and normal upgrades. Switching players requires fresh centering or explicit reuse of that player's saved center at the same camera and playing position. Restoring a hand-setup backup requires centering unless you explicitly reuse its calibration with the same camera and playing positions. Backups include name, personal and effective sensitivity, source software identity, and calibration. Version-2 is the first supported portable format; version-1 sensitivity-only files are rejected. The web footer reports exact software and running firmware identities; older firmware may report unavailable.

Completing all sixteen lessons replaces the lesson panel with the **Glove Master** award. **Start again** restores the lessons.



Setup connection settings and pairing; use HTTPS to enable pairing

Help serves the public manuals, illustrations, and PDFs locally. **This cabinet** shows addresses derived from your current browser connection and public device settings. It never displays the token. The standalone Quick Reference is excluded from the public package; the live cabinet page supplies local details.

Play Checklist

1. Power the RetroPie and PowerGlove Vision Controller; leave the camera connected to the powered hub.
2. Open `http://UNO-Q-NAME.local:8088/dashboard`.
3. Select the active profile on the Dashboard, then confirm the expected profile and a detected hand. The saved startup profile remains on Setup.
4. On first use, or after changing your camera or playing position, select **Calibrate** while holding a comfortable neutral pose. Otherwise reuse the saved calibration.
5. Select **Start controller** when you are ready to arm gesture control.
6. Launch a registered game and confirm its profile code on the matrix. Delivery begins only after RetroArch is running and its short initialization guard ends.
7. Select **Stop controller** before adjusting the camera or leaving the cabinet.
8. Read the shutdown limitation before disconnecting power. **Shutdown** requests a graceful halt, but the tested board restarts; an offline website is not proof that it is safe to unplug.

PowerGlove Vision remembers the player's explicit **Start controller** or **Stop controller** choice across Controller application and system restarts. A remembered Start means **armed**, not unconditional output: controls are sent only during a live registered RetroArch session or after an intentional manual Dashboard profile selection. A registered game renews its session while RetroArch is running, so an Controller application restart can reconnect automatically. Game exit, an unknown game, or an expired session releases all controls and stops delivery without changing the armed preference. **Stop controller** remains sticky until explicitly started again. Install the same release on both devices because this behavior uses a matching Controller worker and RetroPie launch hook.

Vision and the dashboard keep running while output is unarmed or waiting for a game, so setup never generates surprise game inputs. **Shutdown** is different: it halts Linux on the Controller. The tested board automatically restarts; remaining halted is not guaranteed.